

Position-Aware Mask-Energy Normalization for a Finite Masked Twin-Prime Shell Operator

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Abstract

The preceding finite source/operator audit transferred positive operator polarization to higher origins but showed a downward movement of its all-plus minimum floor. We test a predeclared, response-independent position-aware normalization. For each unsigned prime component B_p we form the diagonal mask-energy $G_u = \sum_{p,t} B_p(u,t)^2$ and apply the symmetric congruence $A^\sharp = D_G^{-1/2} A D_G^{-1/2}$. The construction uses no source vector, sign law, or observed response. On three frozen panels and 648 law-level rows, the normalized all-plus minimum drop from the low-origin parent to the higher-origin parent is 0.0262369882, versus 0.0421511462 without normalization, a finite reduction fraction of 0.3775498289. The repair is not uniform: the all-plus mean drop becomes larger, and a fresh mod-4 row is still negatively aligned. We therefore record a finite partial repair and a sharp obstruction, not an asymptotic arithmetic estimate or a twin-prime theorem.

1 Question and scope

TPC-353 attached the finite V59 residual $\beta(t) = \Lambda(t+2) - b^{(2)}(t)$ to a literal two-endpoint divisibility-masked shell operator. TPC-354 moved the same construction to disjoint higher origins. Its positive alignment census survived, while the all-plus minimum and mean decreased relative to the low-origin parent.

The present question is deliberately narrower: can a normalization based only on the position-dependent unsigned mask geometry account for that minimum-floor movement? The answer is tested on the locked TPC-353 panel, the locked TPC-354 panel, and a fresh origins-only panel. Every conclusion below is classified as either an exact finite identity, a declared finite model statement, or a numerically certified finite observation. No statement here supplies a growing bound in X .

2 Finite operator and normalization

Let I be a finite integer interval and let $S_Q = \{p : p \text{ prime}, Q < p \leq 2Q\}$. For exponent s and height H , define the unsigned component

$$B_p(u, t) = 1_{u \neq t} 1_{p \nmid u} 1_{p \nmid t} p \frac{H^{2s}}{(H^2 + (u - t)^2)^s} \left(1_{u \equiv t \pmod{p}} - \frac{1}{p-1} \right). \quad (1)$$

For a predeclared sign law $e = (e_p)_{p \in S_Q}$, the literal operator is

$$A_e = \sum_{p \in S_Q} e_p B_p. \quad (2)$$

The endpoint masks in (1) are part of the definition.

The position-aware geometry energy is

$$G_u = \sum_{p \in S_Q} \sum_{t \in I} B_p(u, t)^2, \quad D_G = \text{diag}(G_u), \quad A_e^\sharp = D_G^{-1/2} A_e D_G^{-1/2}. \quad (3)$$

The square is taken before signs are inserted. Consequently G_u is independent of e , and it also does not use Λ , b , β , or an output vector. On every audited row $G_u > 0$, so the congruence is well defined. It is a finite preconditioner, not a claimed uniformly bounded operator family.

3 Exact finite identities

For either $T = A_e$ or $T = A_e^\sharp$ and $\beta = L - b$, finite bilinearity gives

$$\|T\beta\|_2^2 = \|TL\|_2^2 + \|Tb\|_2^2 - 2\langle TL, Tb \rangle. \quad (4)$$

Writing $E_L = \|TL\|_2^2$, $E_b = \|Tb\|_2^2$ and

$$\kappa_T = \frac{2\langle TL, Tb \rangle}{E_L + E_b}, \quad R_T = \frac{\|T\beta\|_2^2}{E_L + E_b}, \quad (5)$$

we have the exact finite relation $R_T = 1 - \kappa_T$. Cauchy–Schwarz gives

$$\frac{(\sqrt{E_L} - \sqrt{E_b})^2}{E_L + E_b} \leq R_T \leq \frac{(\sqrt{E_L} + \sqrt{E_b})^2}{E_L + E_b}. \quad (6)$$

The diagonal congruence changes the finite geometry on which these identities are evaluated; it does not change their logical status.

4 Frozen protocol and audit

The three panels are shown in Table 1. Counts, shells, exponents, height, source cutoff, source convention, and four sign laws are identical across panels. The fresh panel was fixed before its responses were computed.

Table 1: Frozen three-panel protocol.

panel	origins
low parent (TPC-353)	6001, 8001, 10001
higher parent (TPC-354)	21001, 23001, 25001
fresh holdout	29001, 33001, 37001
source counts	256, 512, 1024
shell anchors	$Q = 24, 54, 80$
kernel exponents	$s = 1, 2$
height / source cutoff	$H = 66 / 50000$
sign laws	all-plus, alternating-index, mod-4, half-split

The Cartesian product has $3 \cdot 3 \cdot 3 \cdot 2 \cdot 4 = 216$ rows per panel, 648 rows in total. The producer uses the finite V59 midpoint convention and accumulates shells in increasing order. A separate checker reconstructs the source and accumulates shells in reverse order; it does not import the producer. Both raw and normalized metrics are compared. An exact rational fourteen-point anchor on [29001, 29014] with shell {5, 7} verifies the unscaled polarization identity, while positivity of the rational geometry diagonal is checked independently. Ten in-memory mutations are required to be rejected by the certificate stress suite.

5 Results

Table 2 records the all-plus minima and means. The normalization reduces the low-to-higher minimum drop, but does not reduce the corresponding mean drop.

Table 2: All-plus output coefficient κ_T by panel.

panel	raw A_e			normalized $A_e^\sharp$		
	min	mean	max	min	mean	max
low	.692912	.895612	.996268	.690971	.899454	.996303
higher	.650760	.874362	.991350	.664734	.874614	.989806
fresh	.654458	.880122	.994391	.664140	.886090	.993322

The raw higher-panel minimum drop is

$$0.69291151430780062 - 0.65076036812307647 = 0.042151146184724153.$$

The normalized drop is

$$0.69097110464200440 - 0.66473411648923819 = 0.026236988152766205,$$

so the descriptive drop-reduction fraction is

$$1 - \frac{0.026236988152766205}{0.042151146184724153} = 0.37754982894688971.$$

On the fresh panel, the normalized minimum is 0.6641398063, only -0.0005943102 below the higher-panel normalized minimum. This is evidence for a finite partial stabilization of the minimum, not a uniform lower-floor statement.

The mean gives the opposing signal. The raw low-to-higher mean drop is 0.021249745559872912, whereas the normalized mean drop is 0.024839744603963321. For the four laws, the fresh panel retains one negative mod-4 row in each metric family: the raw negative row is $(33001, 256, Q = 24, s = 1)$ and the normalized negative row is $(33001, 512, Q = 24, s = 1)$. The higher-panel half-split minimum also decreases under normalization, from 0.0393482615 to 0.0353891510.

Table 3: Alignment census over all 648 rows.

metric	positive	negative	unresolved
raw A_e	647	1	0
normalized $A_e^\sharp$	647	1	0

Thus the normalization improves a selected all-plus minimum statistic while leaving a law-level sign obstruction and a mean-transfer obstruction intact.

6 Interpretation and route status

The reusable structure is a source-independent geometry layer that can be inserted between a literal masked operator and the finite polarization interface. It makes the location of the finite obstruction more explicit: normalizing unsigned endpoint/mask energy can alter floor behavior without selecting a canonical sign law.

The strongest licensed claims are:

- the geometry diagonal and diagonal congruence are exact finite declared constructions;
- polarization and its Cauchy envelope hold exactly for both finite operators;
- the raw/normalized three-panel replay is numerically certified on 648 rows, with independent reverse-shell and mutation controls;
- the all-plus minimum-floor reduction is a finite partial repair, while mean repair and law-uniform alignment are scoped refuted.

No source-uniform arithmetic L^2 estimate, growing geometry bound, canonical sign law, fixed-power credit, Route-B reassembly, or twin-prime conclusion follows. The Session-named official Route-A/Route-B evaluator files are absent from this checkout; the local Bridge-B result is therefore fail-closed fallback evidence only.

7 Reproducibility

The project directory contains the source, independent checker, stress suite, canonical JSON certificate, proof and theorem ledgers, and this manuscript. The top-level README gives the exact normal and optimized commands. The certificate status is

NUMERICALLY_CERTIFIED_FINITE_POSITION_AWARE_MASK_ENERGY_NORMALIZATION_AUDIT.

The arithmetic advance is ‘NO’, fixed-power credit is zero, and the next natural experiment is an adversarial position/origin holdout.

References

- [1] G. H. Hardy and J. E. Littlewood, Some problems of “Partitio Numerorum” III: On the expression of a number as a sum of primes, *Acta Math.* 44 (1923), 1–70.
- [2] G. Greaves, *Sieves in Number Theory*, Springer, 2001.